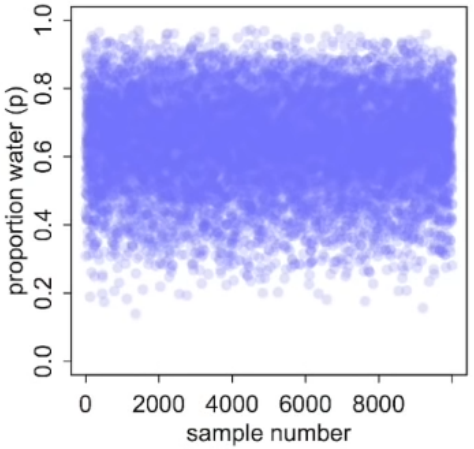
Lecture 8: Sampling from the posterior and from other beasts

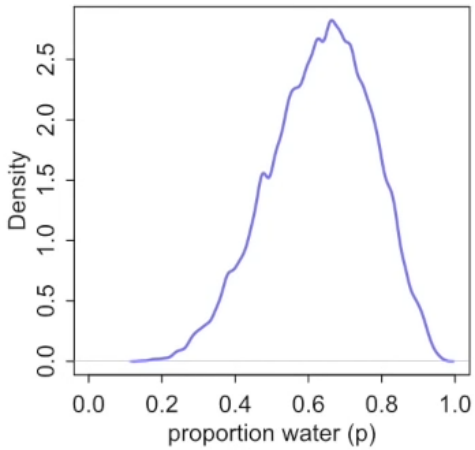
- Very useful to randomly sample from the posterior:
  - Visualise uncertainty
  - Compute intervals
  - Simulate observations
- Only by (relative) counts!
- Moreover, when we will encounter MCMC: samples from the posterior as an output
- Easier to think of than probabilities
- Transform (hard) calculus problems into (easy) summary data problems.

Sample from the posterior recipe

- Compute a posterior. Most of the time an approximate posterior.
- Sample with replacement form the posterior.
- Compute summaries from the sampled valued to get quantitative answers to your question(s).

```
samples <- sample(p_grid, prob = posterior,
    size = 10000, replace = TRUE)
plot(samples)
dens(samples)
```





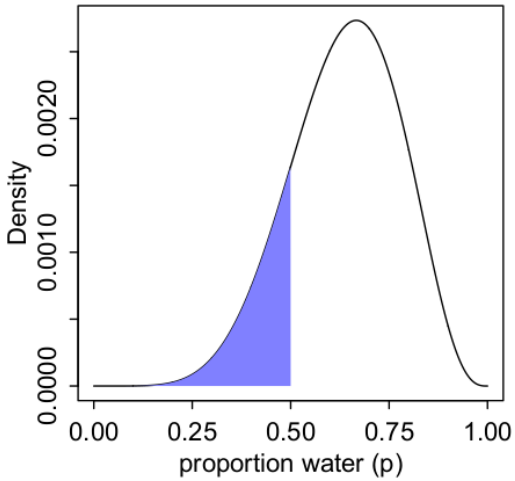
Summaries as features of posterior samples

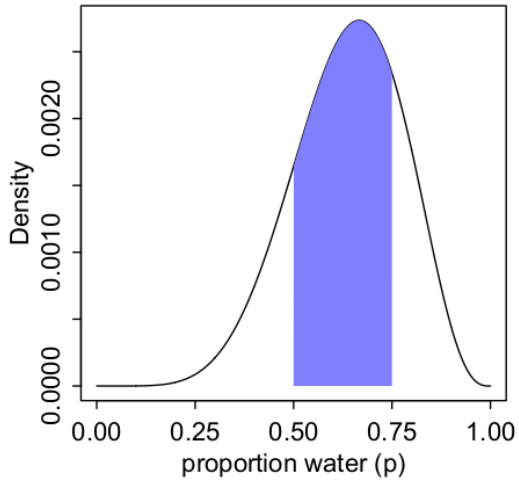
- Useful summary tasks depend on the question at hand. You decide among the many possibilities:
  - Intervals 1: how much of the posterior probability is above/below/between specific (parameter) values?
  - Intervals 2: which parameters values contain 50%/80%/99% of the posterior probability. NOT confidence interval with their fictional repeated experiments.
  - Point estimates: mean, median, mode of the posterior? Value that minimises a posterior loss (see SR)?
  - Etc.

See: "Counting with `sum()`" box in SR Chapter 3. Also use of `quantile()`

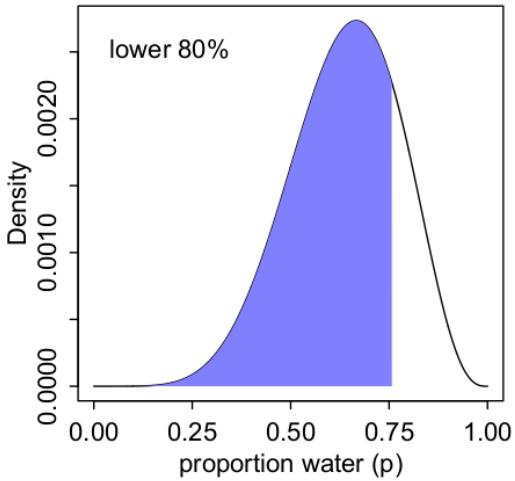
Posterior summary interval examples 1

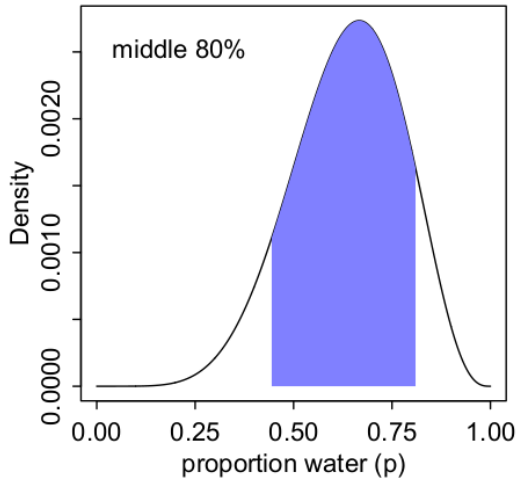
Intervals of defined boundary ask *how much mass?*





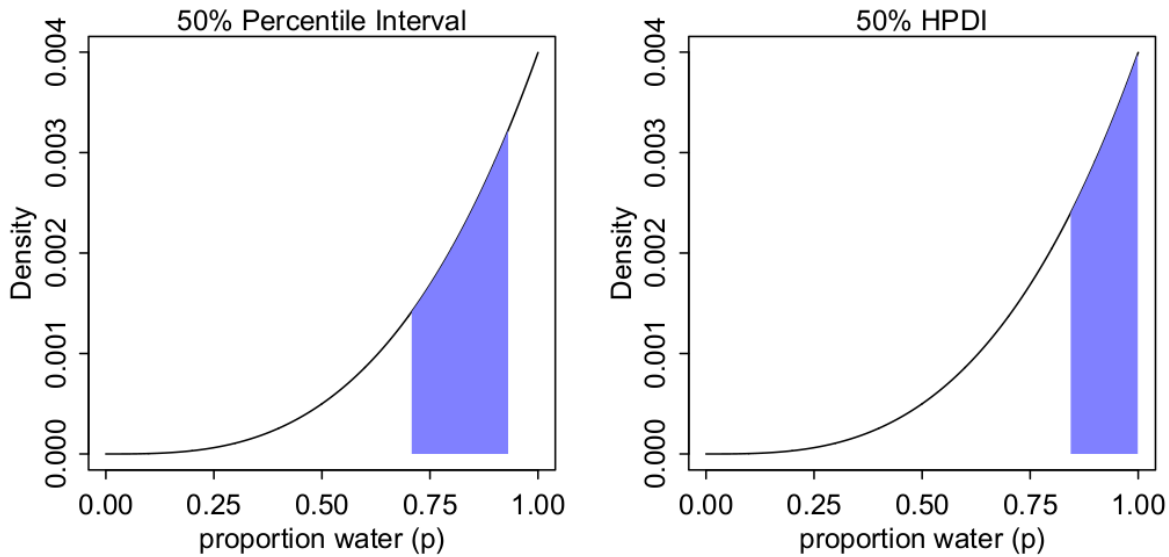
Intervals of defined mass ask *which values?*





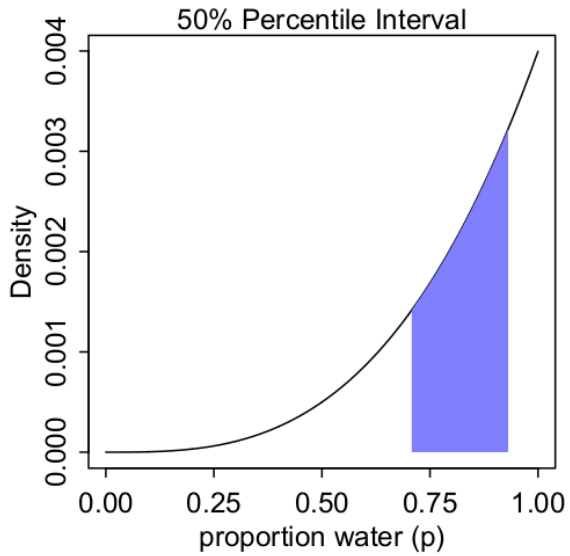
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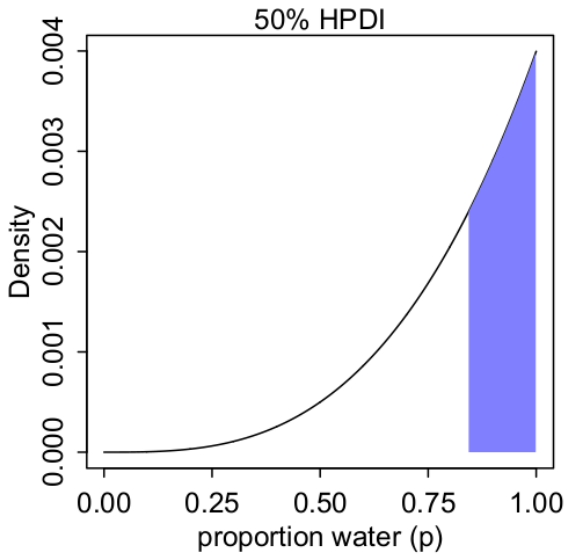
Posterior summary interval examples 2



- Percentile intervals (PI): equal area in each tail.
- Highest Posterior Density Interval (HPDI): narrowest interval containing the defined mass.

Posterior summary interval examples 2






```
PI(samples, prob = 0.5)
      25%       75%
0.7037037 0.9329329
```

```
HDPI(samples, prob = 0.5)
lower 25% upper 75%
0.8418418 1.0000000
```

Repeat: the entire posterior matters

- Usually we don't want/need point estimate.
- More information in the entire posterior.
- The best/appropriate/most appropriate point depends upon the purpose.
- Using a point estimate only can mask uncertainty!
- Term 'credible' interval frequent in Bayesian analysis. Not ideal, shall we grant confidence or credibility to the interval? $\rightarrow$ 'compatibility' (with the model and the data) interval. Up to you to believe the model and the data

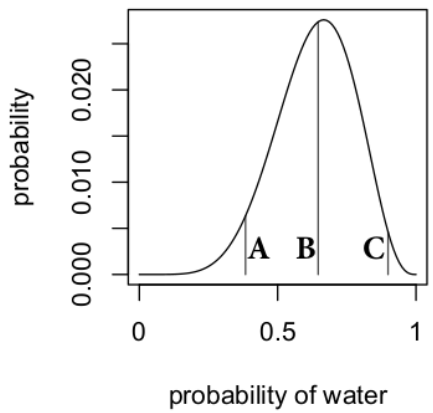
Sampling to simulate predictions

Predictive checks

- Posterior probability is rarely enough.
- To judge the quality of a (the 'best') model, we may want to judge it based on its predictions. Good enough?
- Feedback on model assumptions.
- Predictive checks: we can use samples from the (parameter) posterior(s) to simulate observations. And then compare these simulations to actual data. Or to another reference.

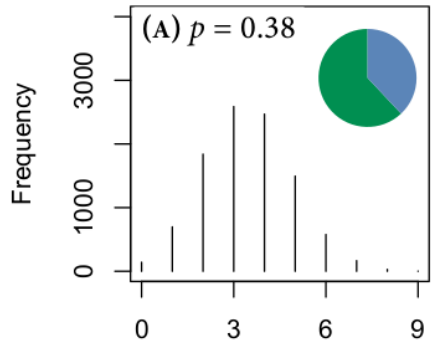
Sampling Water vs Land on Earth

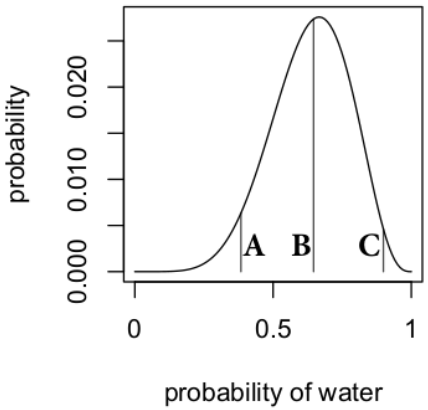
from 3 scenarios



Sampling Water vs Land on Earth

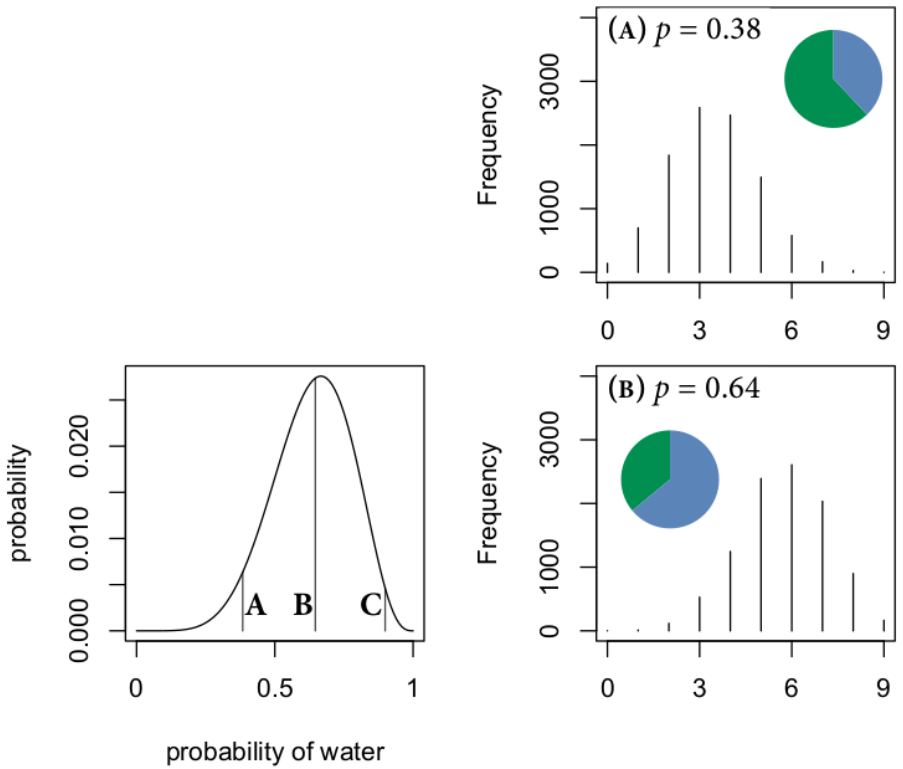
from 3 scenarios





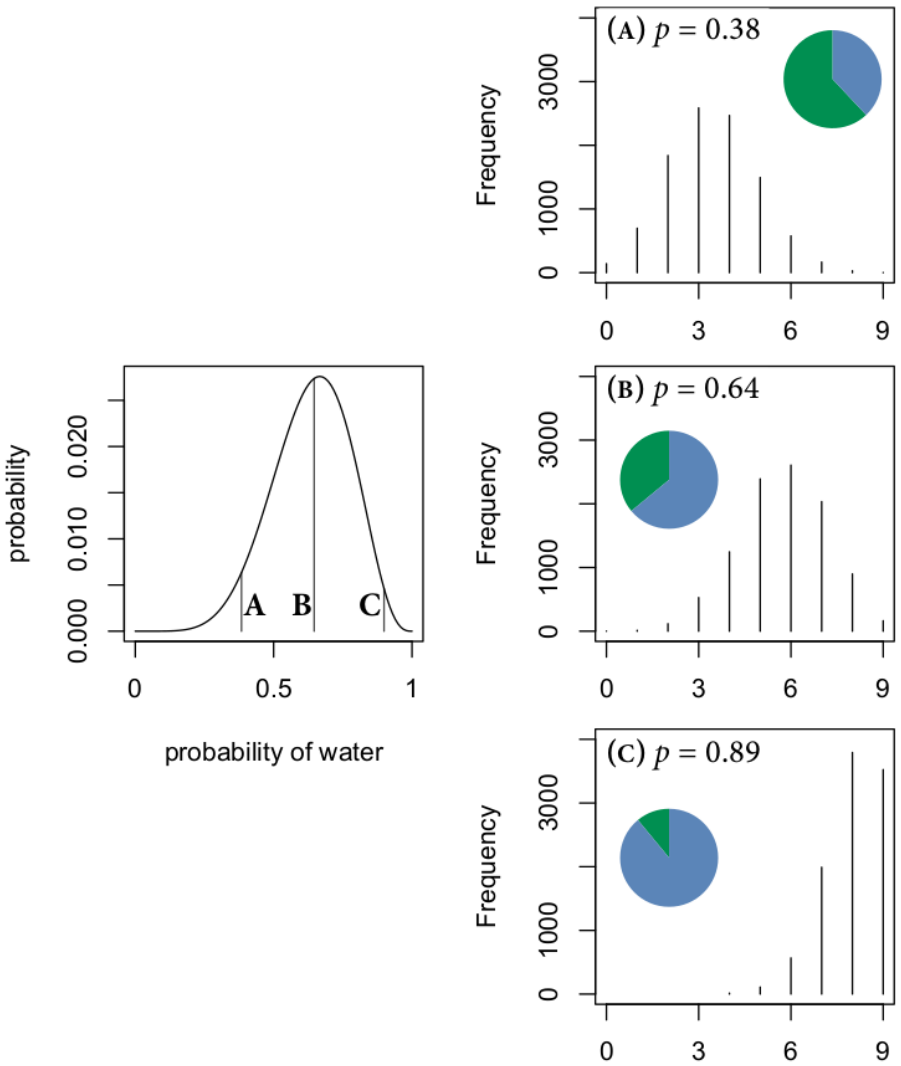
Sampling Water vs Land on Earth

from 3 scenarios



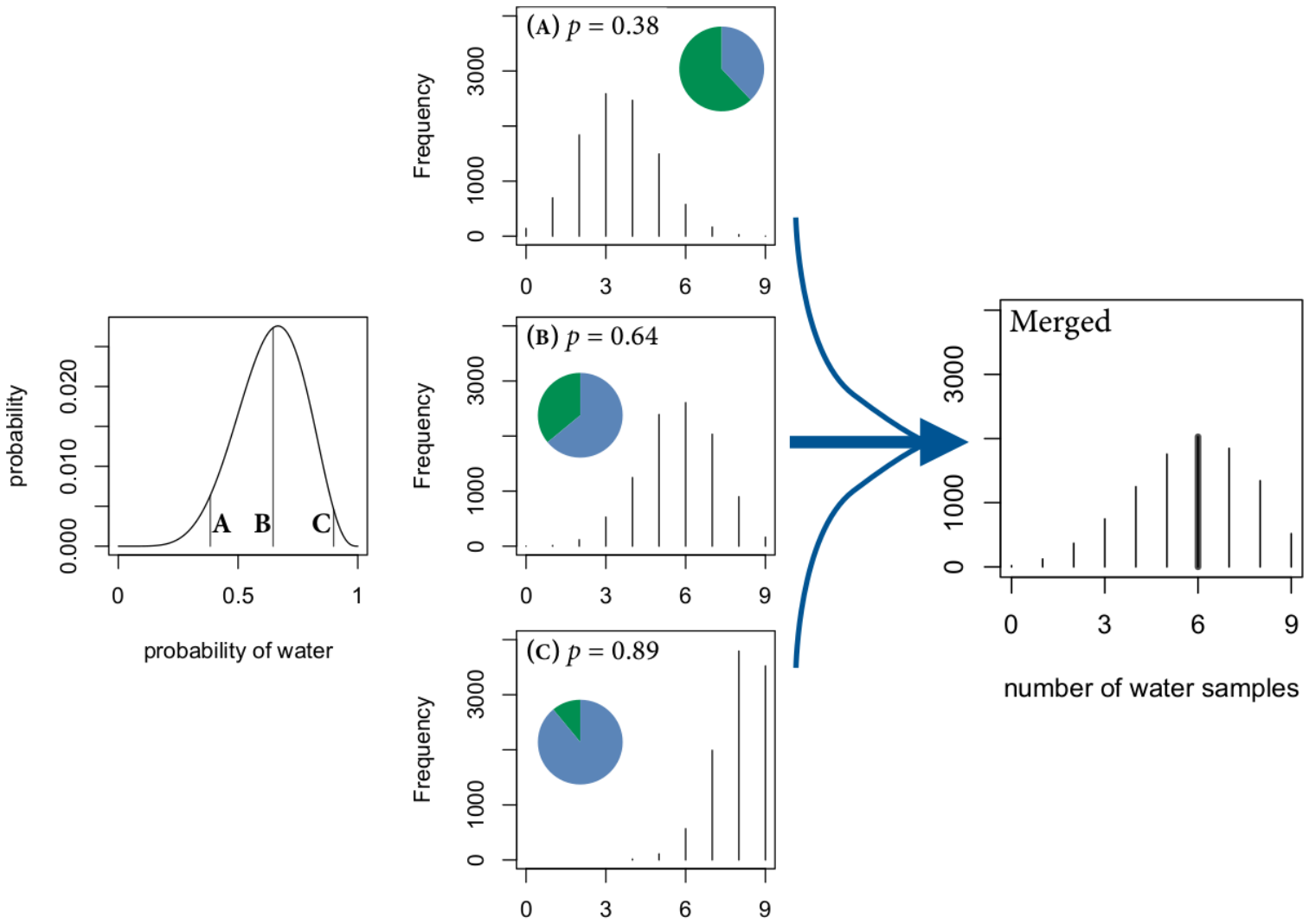
Sampling Water vs Land on Earth

from 3 scenarios



Sampling Water vs Land on Earth

from 3 scenarios



Sampling Water vs Land on Earth

In one line of code

R Snippet

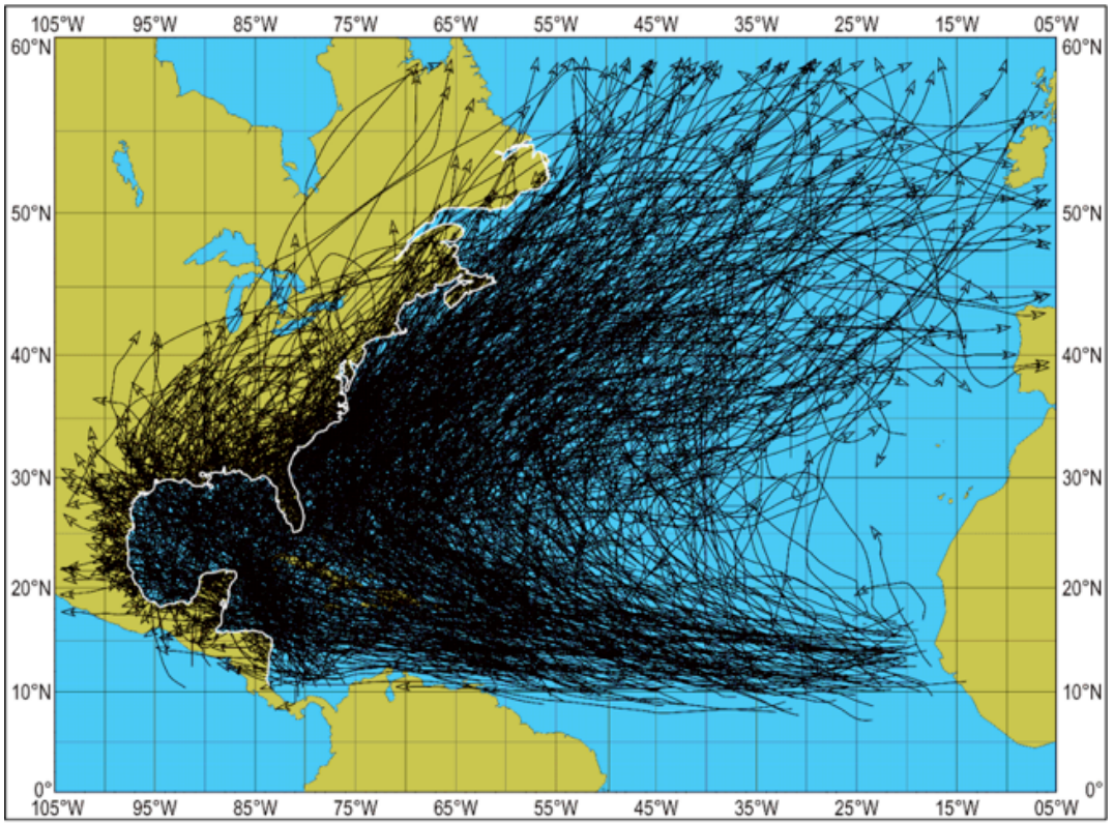
```
sim.water.data <- rbinom(10000, size = 9, prob = samples)
```

- This is called the *posterior predictive distribution*.

- It may be (slightly) more complex when model generating the data are more complex. E.g. if more parameters. But the idea stays the same.

Simulating trajectories

Tropical storms and hurricanes in the North Atlantic Ocean



- Compare it to historical data? Do predictions now and compare it to the next year (or 5-10 years) events?

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Predictive checks

- Resemble significance tests. But it's not.
- No universal best way to evaluate adequacy of a model from its predictions.
  Examples: could look at longest W/L runs (≥ 9). The number of switches (≥ 8)
- Do not work at the religious 5%. Unless you have good reason to do so.
- Good predictive checks depend upon purpose and...your creative imagination. Look for aspects of the data not well explained by the model: a path towards model comprehension, revision and improvement.
- Be aware of (i) observation uncertainty (even when the parameter(s) are fixed), (ii) there is uncertainty about the parameter(s) too (why not about the generating model?!). We 'propagate' these uncertainties.

Quick summary

- We introduced basic procedures to manipulate `samples` from the posterior distributions.
- As a way to check the distribution. Its implications on parameter values. Making data summaries (counts) instead of complex integral calculus.
- We can produce intervals (PI, HPDI), point estimates and predictive values to be checked.
- These are the implications of the posterior: the posterior predictive distribution combine uncertainty about the parameter(s) with uncertainty inherent to the data (outcome) generating process (in the likelihood part). At least we can check that our implementation works.
- More difficult task: once the small-world part is validated, how can it be useful to describe the large-world?
- End of the novelty about simulations part...until MCMC. And all our checks of course!